

Les progressions d'aprenentatge a l'ensenyament de les ciències: què són, per a què serveixen i exemples

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Universitat
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Barcelona

Curs 2025-2026

Les 4 dimensions d'un profe



DOCÈNCIA

L'acció docent a les aules de secundària.

cesire*
cdec

centre de documentació
i experimentació
en ciències

INNOVACIÓ/TRANSFERENCIA de la RECERCA

Elaboració de nous materials didàctics.



**Universitat
Pompeu Fabra
Barcelona**

FORMACIÓ INICIAL I PERMANENT

Compartir experiències

UAB
Universitat Autònoma
de Barcelona



RECERCA EDUCATIVA

Millorar les teories sobre
l'aprenentatge i l'ensenyament

El repte d'un profe de ciències és

Que **TOT** l'alumnat assoleixi el **nivell** de competència científica més elevat que sigui possible, d'acord amb les seves circumstàncies.



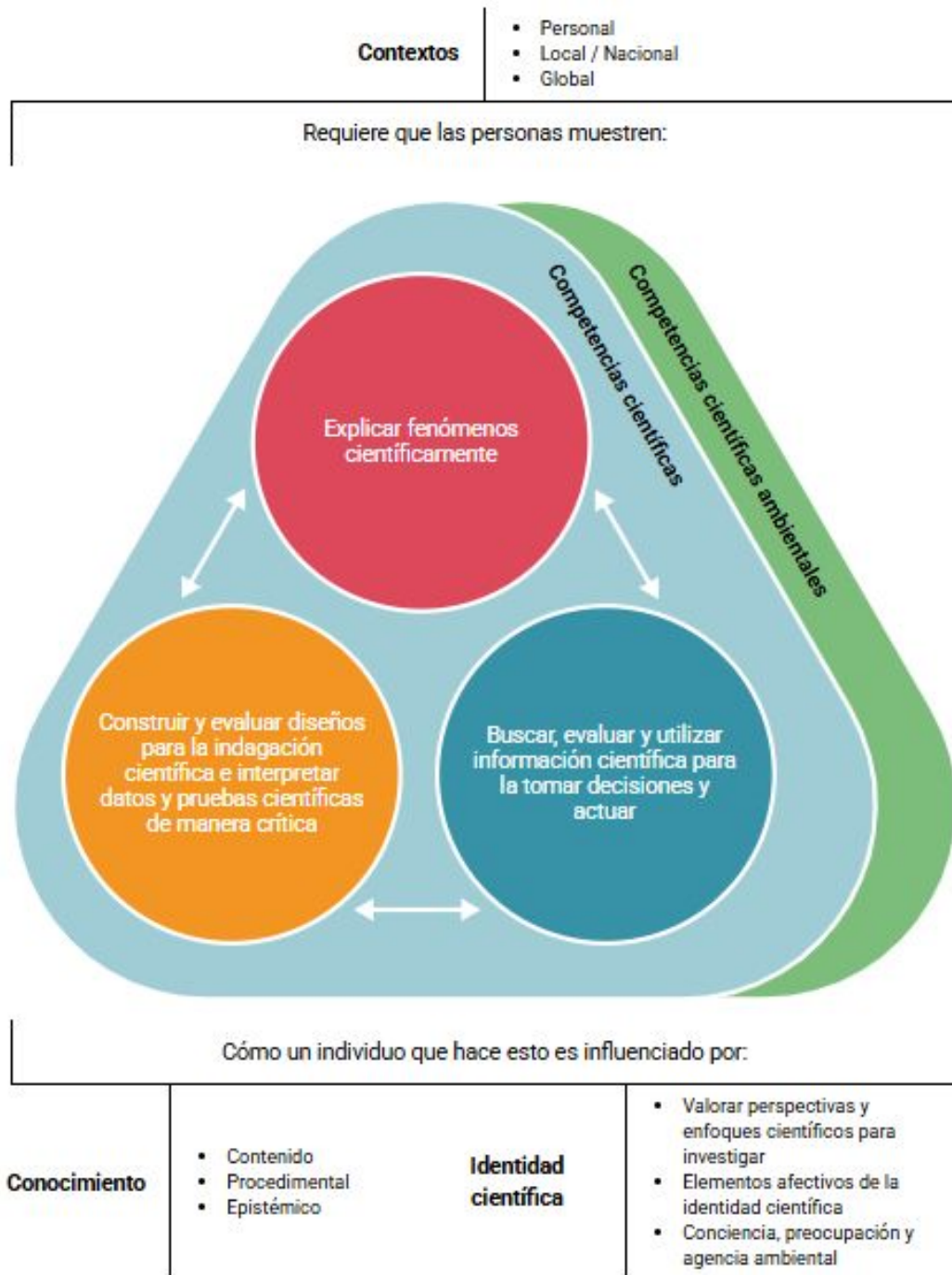
Aleshores, sorgeix la necessitat de consensuar significats:

- Què vol dir nivell ?
- Què vol dir competència científica?

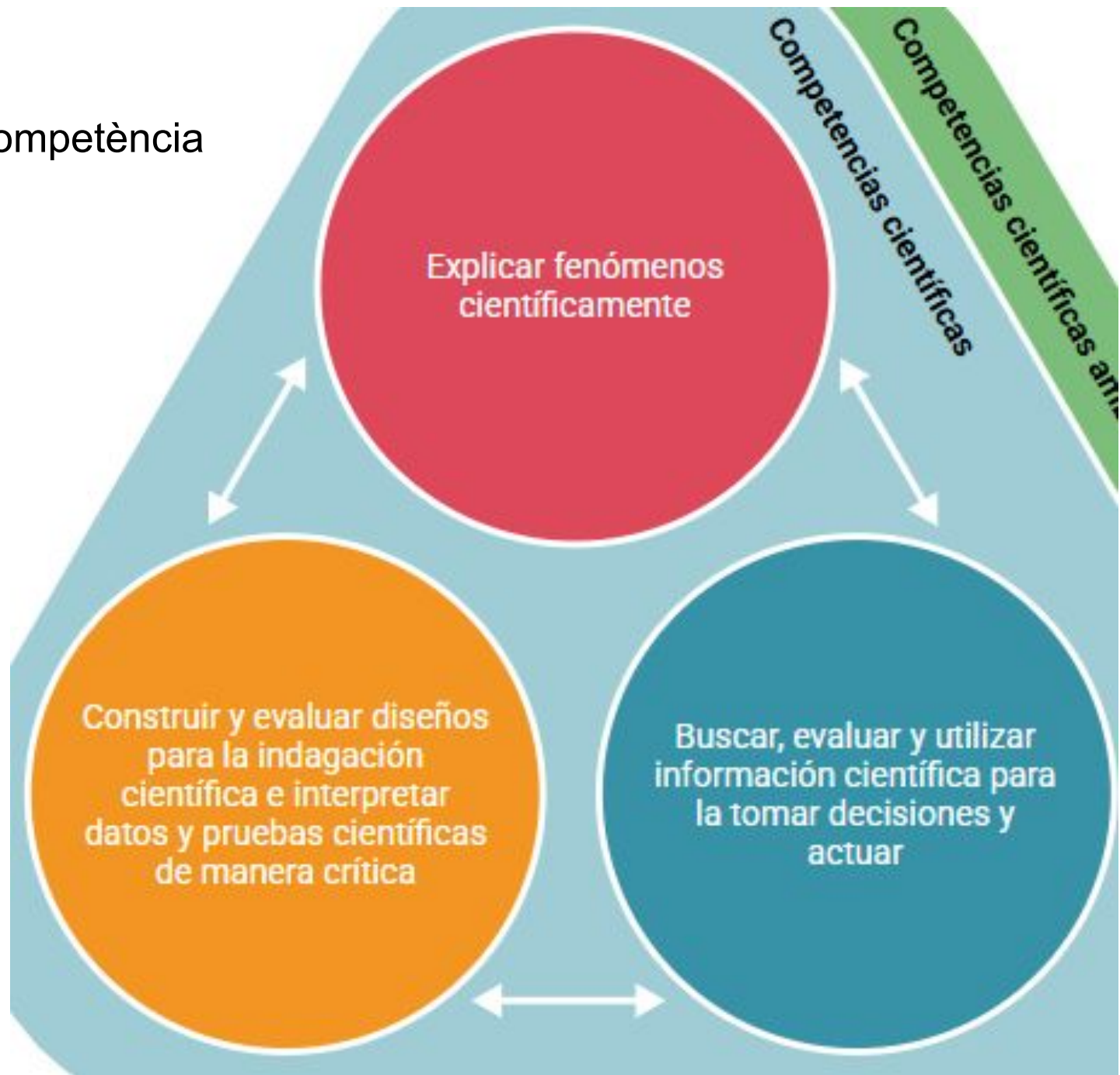
Marc teòric PISA 2025:

- 3 competències
- 3 tipus coneixements
- 3 tipus contextos
- Identitat científica

Font:
https://pisa-framework.oecd.org/science-2025/per_spa/



- Què vol dir competència científica?



Què més diu PISA?



Les competències es fan servir en contextos, situacions del món real:

- **Classificació dels contextos, El més reals possibles, Interdisciplinarietat**
- **No s'avaluen contextos, Diferències culturals pels ctxtos, Important varietat de ctxtos**

	Personal	Local/National	Global
Health & Disease	Maintenance of health, accidents, nutrition, vaccination	Control of disease, social transmission, food choices, obesity, community health	Pandemics, food Security, healthy Lifestyles
Natural Resources	Personal consumption of materials, types of food, and energy. Consuming locally produced foods. Choosing non-dairy and vegetarian diets	Maintenance of human populations, quality of life, security, production and distribution of food, energy supply. Environmental impact of mining and resource extraction. Production of renewable energy	Renewable and non-renewable sources of energy, natural systems, population growth, sustainable use of species and land. Biodiversity and its value
Environmental Impacts & Climate Change	Sustainable practices of recycling and reduction of resource use.	Population distribution, waste management, environmental impact. Use of regenerative agriculture	Environmental sustainability, management of pollution and air quality, loss of soil/biomass. Mass extinction of species. Ocean Acidification
Hazards	Risk assessments of lifestyle choices.	Rapid changes [e.g. earthquakes, severe weather], slow and progressive changes [e.g. coastal erosion, sedimentation], risk assessment. Facial recognition	Threats posed by Climate change, impact of modern communication, energy and its production e.g. fracking, nuclear, gas
Contemporary Scientific and Technological Advances and Challenges	Scientific aspects of the use of new technologies e.g. gene editing, virtual reality and	New materials, devices and processes, genetic modifications, health technology, transport, use of artificial intelligence	Exploration of space, origin and structure of the Universe

- Què vol dir nivell? PISA defineix 8 nivells de competència científica

Nivell més alt, 6	At level 6, working in unfamiliar contexts, students can draw on a range of scientific ideas of high demand from different disciplines to build models, consider their limitations, and use those models to construct or evaluate scientific explanations of complex phenomena. They can apply those explanations to make predictions not only about the phenomena but also about potential future developments or implications for society. Students can identify and explain the purposes of particular enquiries of different types, and which question they are answering. They can apply epistemic and procedural knowledge to evaluate competing designs of complex enquiries such as experiments, field studies or simulations and justify their choices of design. They can transform data from one representation to another and correctly interpret more complex data sets. Students can evaluate the interpretation of data sets drawing on procedural and epistemic knowledge to make reasoned judgements about their accuracy and precision. Drawing on multiple sources of information of high cognitive demand, containing both textual and graphical information, students can identify those sources which are most trustworthy based on one or more scientific criteria or more sophisticated fact-checking procedures. They can provide a justification for their choice drawing on content, procedural or epistemic knowledge of science, and/or social, ethical or economic considerations. In addition, they are able to identify flaws in sources of scientific information – either in their trustworthiness, their use of data, or in the arguments from the evidence. Based on their evaluation, they can provide justifications considering multiple issues for possible decisions and actions.
Nivell més baix, 1c	At level 1c, in everyday personal contexts, students can recognise an explanation of a common macroscopic phenomenon communicated in everyday language by recalling elements of everyday scientific information or observations at the lowest level of cognitive demand. Given a simple question, they can recognise a single source of scientific information that might be relevant. They are able to select which is better of two interpretations of a simple set of data.

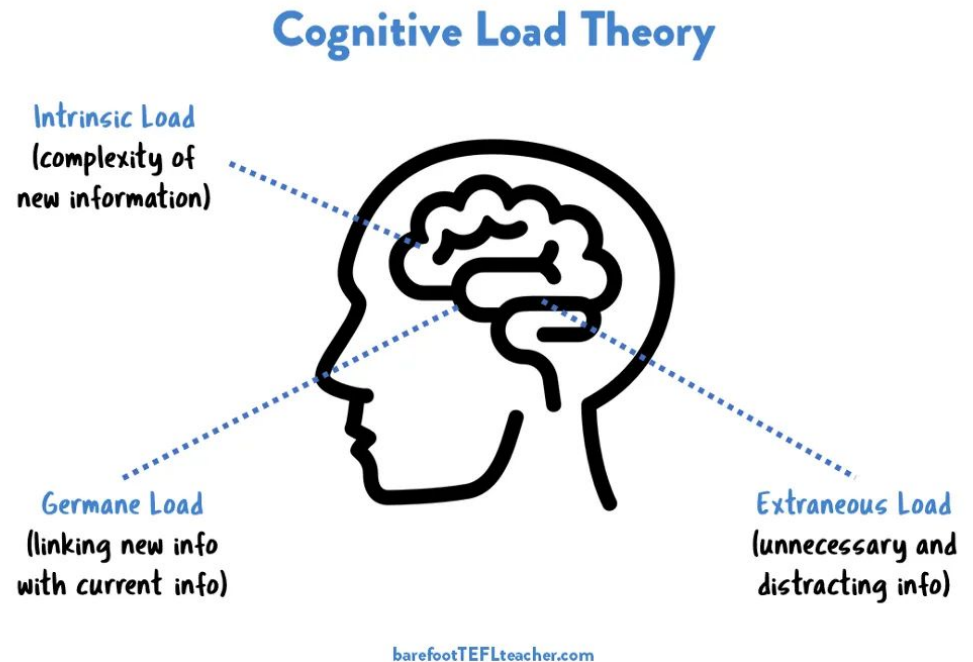
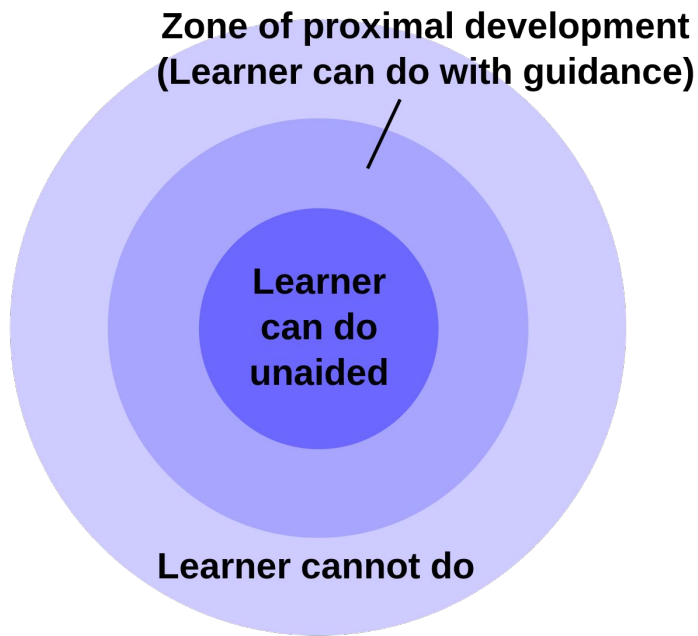
Els nivells de competència depenen dels 4 factors que determinen la demanda cognitiva d'una tasca:



- a. El nombre d'elements de coneixement i el grau de complexitat que demana cada ítem.
- b. El grau de familiaritat i coneixement previ que els estudiants poden tenir del coneixement del contingut, del coneixement epistemològic i del coneixement procedimental.
- c. Quina operació cognitiva precisa l'ítem, per exemple, recordar, analitzar, avaluar.
- d. La mesura en què la resposta donada és dependent d'un model o d'idees abstractes.

Dues teories interessants relacionades:

- La zona de desenvolupament proper
- Teoria de la càrrega cognitiva



Resum en 1 minut

Resum en 1 minut

Com podem desenvolupar la competència científica?

**IDEES
ALTERNATIVES
de l'alumnat**

Progressió d'aprenentatge

**IDEES CLAU de
models teòrics
de la ciència
actual**



**Comunicar
multimodalment
Llengua i TIC**



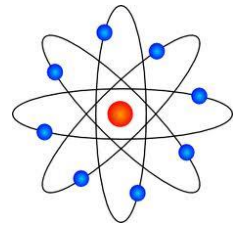
**Cooperar,
treball en equip**

**Activitats
d'ensenyament
i aprenentatge**

Contextualitzar



**Modelitzar amb
experiments
(dins i fora
de l'aula)**



**Indagar,
naturalesa de la ciència**

**Regulació
metacognitiva**



Una analogia del concepte de progressió d'aprenentatge: Anar en bici



Adaptació de la Idea de Digna Couso, LIEC Octubre 2015

Activitat 1. A quin nivell esteu?

Contesteu la pregunta amb el vostre coneixement. Penseu també què respondria un alumne de 1r, 2n, 3r, 4t d'ESO o de 1r o 2n BTX.

Pel que fa als àtoms, quina és la resposta correcta?

- a) Els àtoms són esferes amb els electrons enganxats per la superfície exterior.**
- b) Els àtoms tenen electrons que es mouen seguint orbites al voltant del nucli com els planetes del sistema solar**
- c) Els àtoms tenen electrons que descriuen trajectòries a l'espai definit pels orbitals**
- d) Els àtoms són esferes indivisibles**
- e) Cap de les anteriors**

Activitat 1. A quin nivell esteu?

Discussió de les respostes:

- a) Els àtoms són esferes amb els electrons enganxats per la superfície exterior.

Model de Thompson

- b) Els àtoms tenen electrons que es mouen seguint orbites al voltant del nucli com els planetes del sistema solar

Model de Rutherford

- c) Els àtoms tenen electrons que descriuen trajectòries a l'espai definit pels orbitals

Model quàntic amb un error, segons el principi d'incertesa de Heisenberg no podem tenir trajectòries dels electrons.

- d) Els àtoms són esferes indivisibles

Model de Dalton

- e) Cap de les anteriors

Resposta correcta - Segurament alguns o molts heu votat la b o la c.

Les progressions poden coincidir total o parcialment amb la història de la ciència



DALTON
[1803]



THOMSON
[1904]
Cargas positivas
y negativas



RUTHERFORD
[1911]
El núcleo

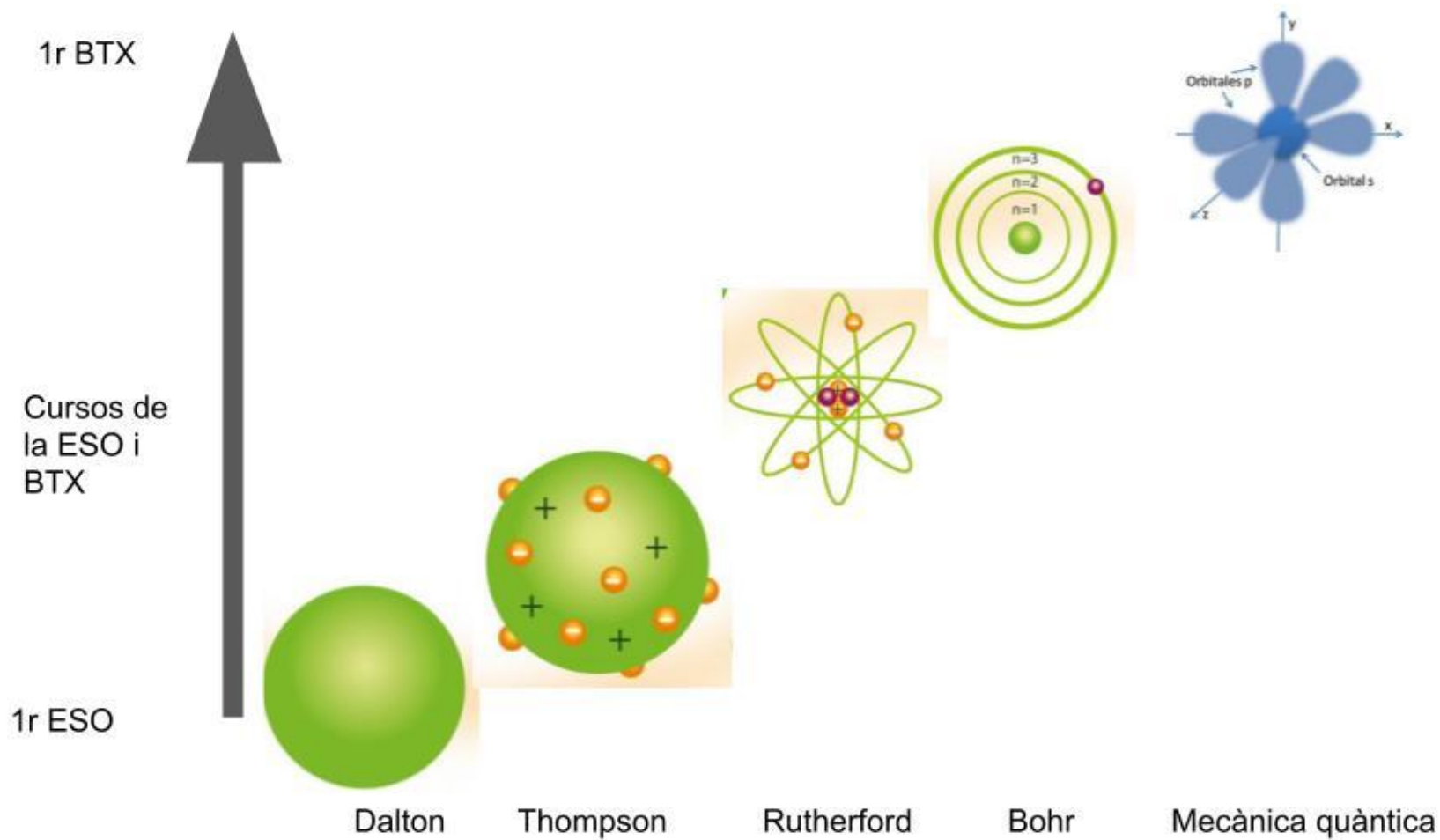


BOHR
[1913]
Niveles de
energía



SCHRÖDINGER
[1926]
Nube de
electrones

2012 - Mariano Díaz Valdecantos



Algunes definicions

Les progressions d'aprenentatge són:

descriptions of the successively more sophisticated ways of thinking about a topic that can follow one another as children learn about a topic over a broad span of time.

National Research Council, USA, 2007

hypothetical pathways for students' science learning.

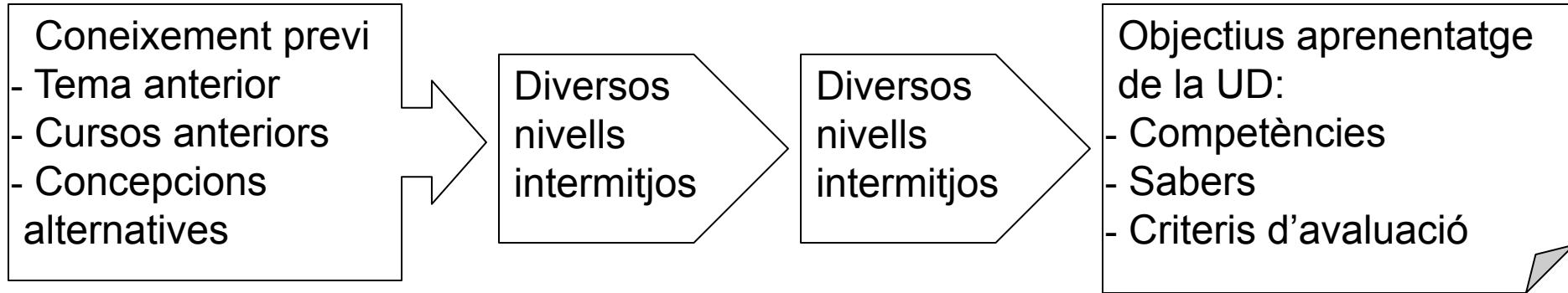
Dusch et al. 2011

modelos curriculares que describen con palabras y ejemplos secuencias de aprendizaje a través de las cuales se propone que los estudiantes prosperen hacia un entendimiento de carácter más experto. Estas progresiones presentan formas de pensamiento sucesivamente más complejas que es probable que los estudiantes sigan conforme aprenden sobre un tema. Las progresiones de aprendizaje se desarrollan alrededor de las grandes ideas de un dominio de la ciencia.

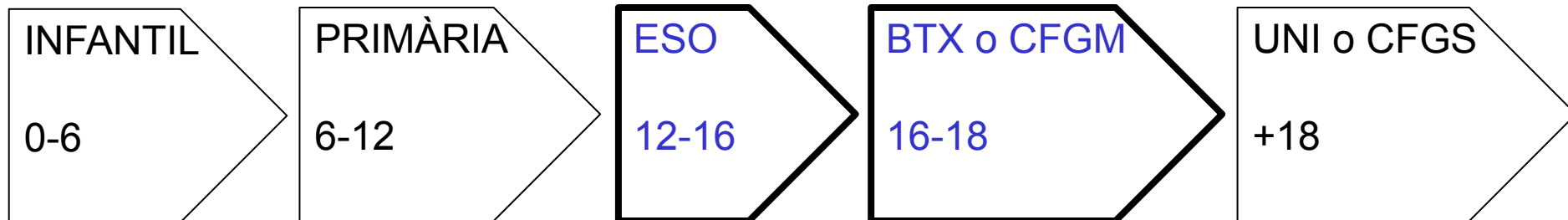
Talanquer 2014

On tenir-les presents

En una unitat didàctica:



En diverses etapes educatives: (A 2n ESO no és el 1r cop que fan física!!)



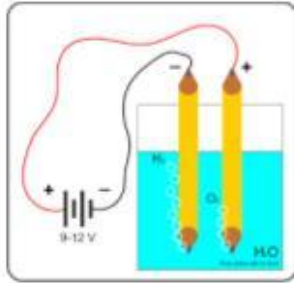
Progressió Química a secundària

Model teòric	1r - 2n ESO	3r ESO	4t ESO	1r BAT	2n BAT
Substància (macro)	m, V, d, mescles	Taula Per. Element VS compost	Propietats i tipus subst	Gasos $PV = nRT$	Diagrama de fases
Substància (submicro)	Model partícules (s), (l), (g)	Ions, àt, molècules, estr. gegant	Enllaç Q. Mol, Estr i prop	Enllaços Intermolec.	Teories de veloc. RQs Colisions i estat de transició
Substància (subnano)	Indivisible Molecula	p+, n, e- Isòtops, ions Massa i volum	Capes d'e- Emissió de llum	Config.elec Espectres	Interac. rad-materia RMN, IR
Canvi químic (del macro al nano)	Noves subst = noves prop	Exemple senzill. Reorg.àtoms 3 nivells Massa	+Exemples Trenc. enll. Energia Velocitat	Estequio. Reactiu limitant Rendiment	Equilibri Q. Entalpia Entropia

Progresió “Canvi químic” a 2aria

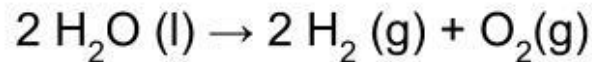
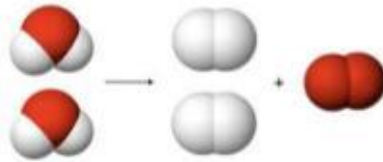
Caamaño y Marchán-Carvajal (Gener 2021) Alambique

Cursos de
la ESO



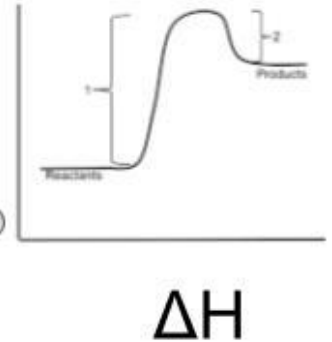
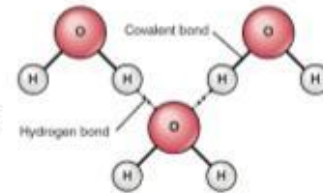
Agua líquida → Hidrógeno gas + oxígeno gas

MACRO



SUBMICRO

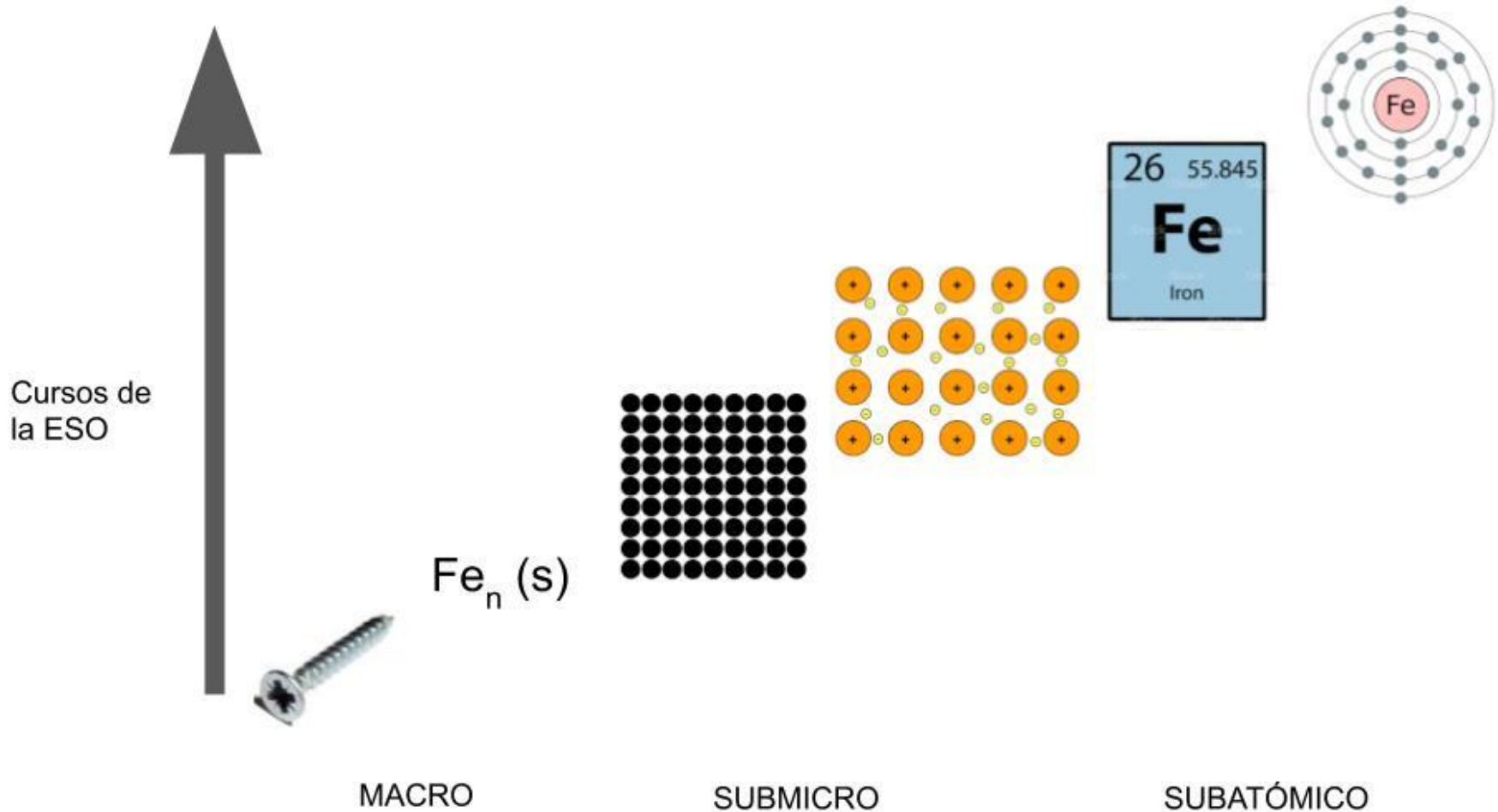
dibujo escala
multi/meso?



SUBATÓMICO

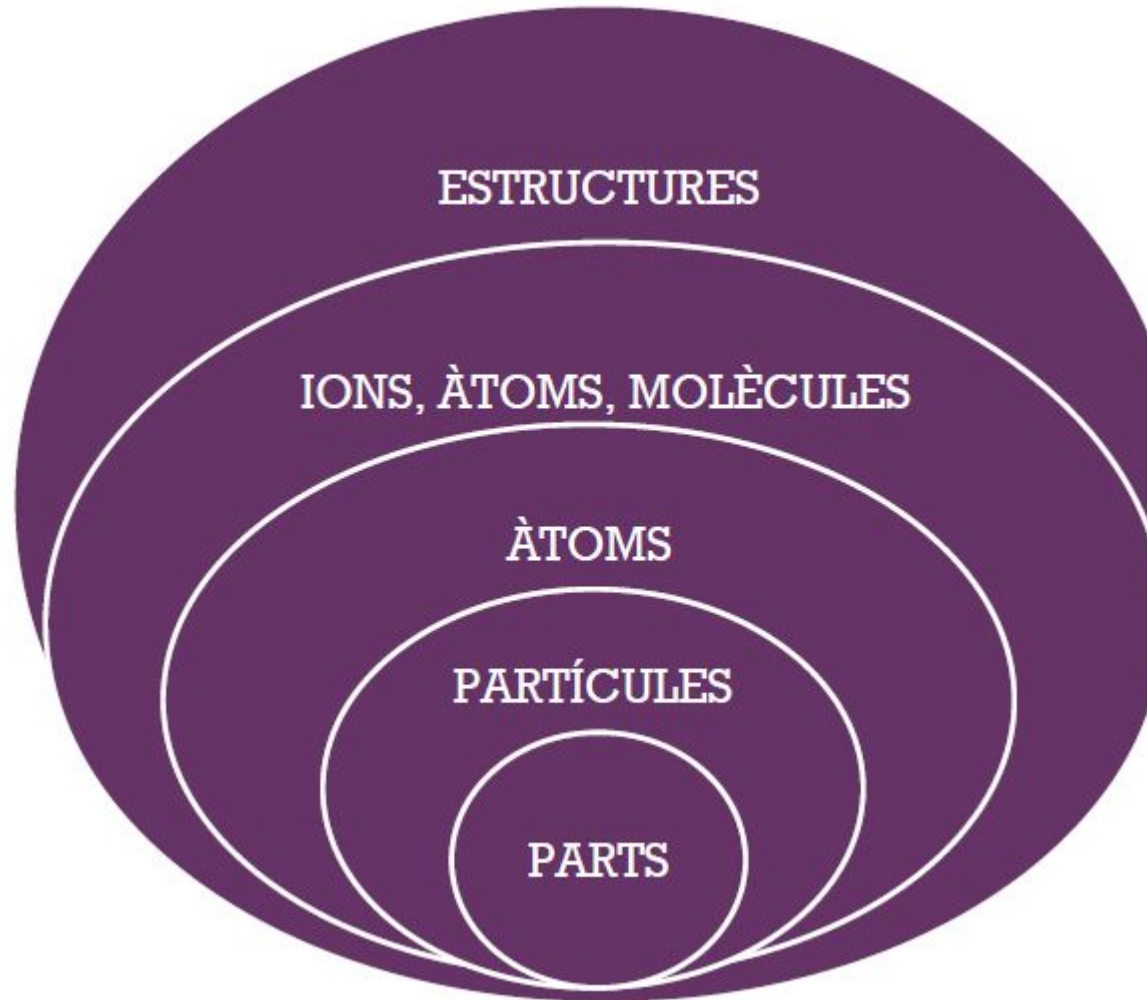
Progresió “Substància” a 2aria

Caamaño y Marchán-Carvajal (Gener 2021) Alambique



Visió discontinua de la matèria dsd Primària

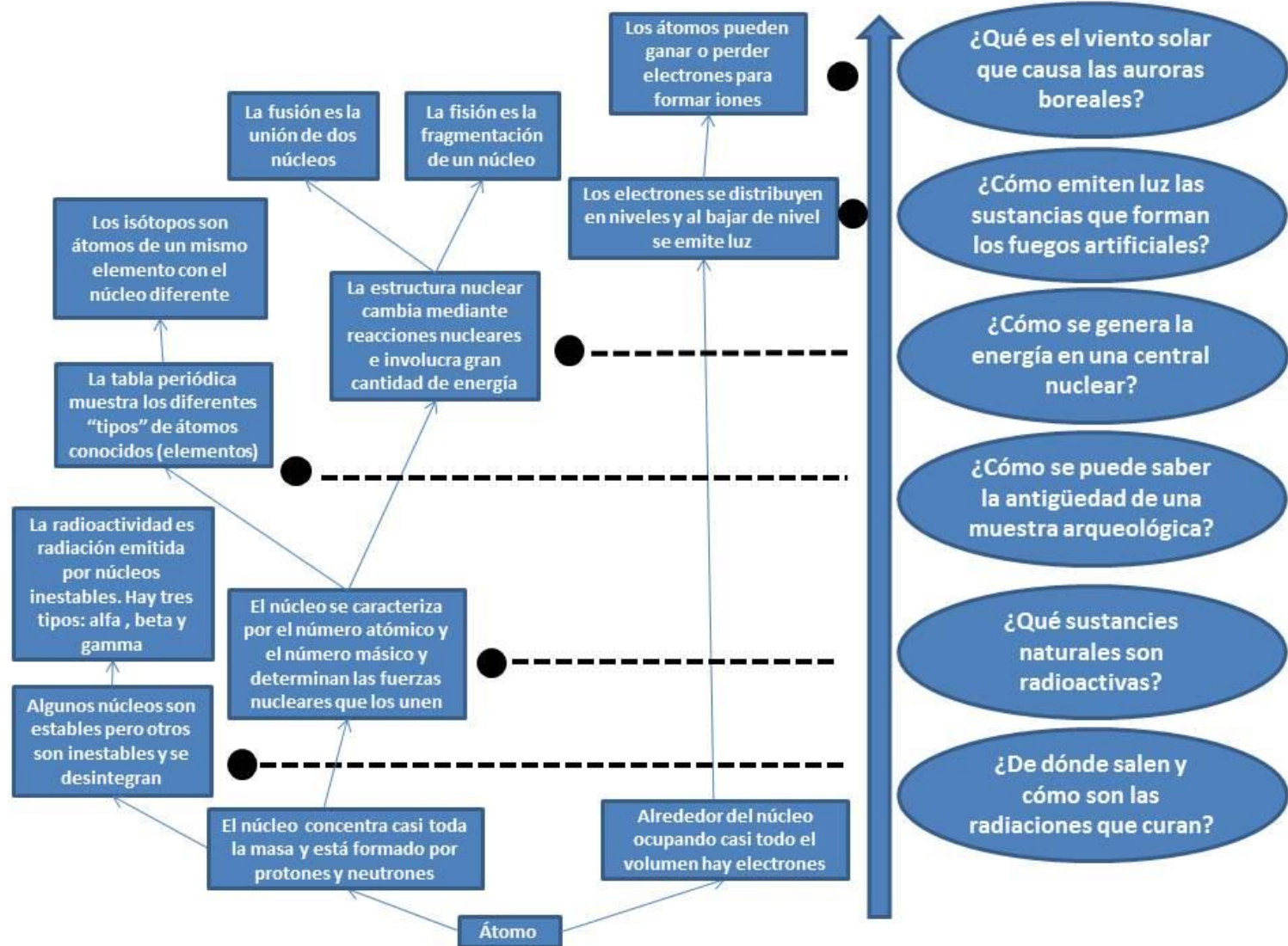
(Marchán-Carvajal i Sanmartí, 2016) Revista EduQ del IEC



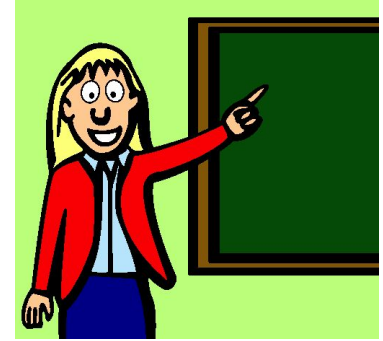
Progressió en una UD

(Marchán-Carvajal i Sanmartí, 2015) Educación química

Cuestiones guía en contexto = obj. d'aprenentatge



Per a què li serveixen a un profe les progressions d'aprenentatge?



- Ajuden a definir els “estándards” que han d’assolir els alumnes a cada etapa
- Proporcionen punts de referència per a elaborar criteris d’avaluació
- Contribueixen a dissenyar els currículums oficials dels marcs legals

Corcoran et al. 2009

- Tenir més presents les concepcions alternatives per ajudar l’alumnat a superar-les
- Promouen el coneixement didàctic del contingut (PCK en anglès) dels professors
- Guia per al disseny de seqüències didàctiques (i elaboració de llibres de text)
- Selecció de les idees clau
- Disseny d’instruments d’avaluació com rúbriques
- Tenir més present el coneixement de cursos anteriors y els fonaments que s’han de donar, per a cursos posteriors
- Estratègies d’atenció a la diversitat (continguts mínims, activitats d’ampliació...)
- Adaptació del currículum oficial a cada centre (acords de departament)

Durant el curs 2024-2025, el departament d'educació va crear unes equips de treball per a elaborar orientacions per al desplegament del currículum, les trobareu aquí:

<https://projectes.xtec.cat/nou-curriculum/orientacions-curriculars/>

A continuació teniu altres exemples de progressions d'aprenentatge. Mireu les que us interessin. En teniu més en un arxiu pdf a l'aula global.

Example 1: Terra-sol-luna

(Dusch et al 2011)

Level	Description
5 8 th grade	Student is able to put the motions of the Earth and Moon into a complete description of motion in the Solar System which explains: - the day/night cycle - the phases of the Moon (including the illumination of the Moon by the Sun) - the seasons
4 5 th grade	Student is able to coordinate apparent and actual motion of objects in the sky. Student knows that - the Earth is both orbiting the Sun and rotating on its axis - the Earth orbits the Sun once per year - the Earth rotates on its axis once per day, causing the day/night cycle and appearance that the Sun moves across the sky - the Moon orbits the Earth once every 28 days, producing the phases of the Moon COMMON ERROR: Seasons are caused by the changing distance between the Earth and the Sun. COMMON ERROR: The phases of the Moon are caused by a shadow of the planets, the Sun, or the Earth falling on the Moon.
3	Student knows that - the Earth orbits the Sun - the Moon orbits the Earth - the Earth rotates on its axis However, student has not put this knowledge together with an understanding of apparent motion to form explanations and may not recognize that the Earth is both rotating and orbiting simultaneously. COMMON ERROR: It gets dark at night because the Earth goes around the Sun once a day.
2	Student recognizes that: - the Sun appears to move across the sky every day - the observable shape of the Moon changes every 28 days Student may believe that the Sun moves around the Earth. COMMON ERROR: All motion in the sky is due to the Earth spinning on its axis. COMMON ERROR: The Sun travels around the Earth. COMMON ERROR: It gets dark at night because the Sun goes around the Earth once a day. COMMON ERROR: The Earth is the center of the universe.
1	Student does not recognize the systematic nature of appearance of objects in the sky. Students may not recognize that the earth is spherical. COMMON ERROR: It gets dark at night because something (e.g., clouds, the atmosphere, "darkness") covers the Sun. COMMON ERROR: The phases of the Moon are caused by clouds covering the Moon. COMMON ERROR: The Sun goes below the Earth at night.
0	No evidence or off-track

Example 2: Models científicos

(Schwartz et al 2009)

Table 3
A learning progression for understanding models as generative tools for predicting and explaining

Level	Performances
4	Students <i>construct and use models spontaneously</i> in a range of domains to help their own thinking. Students <i>consider how the world could behave according to various models</i> . Students construct and use models to generate new questions about the behavior or existence of phenomena.
3	<i>Students construct and use multiple models to explain and predict more aspects of a group of related phenomena.</i> Students view models as tools that can support their thinking about existing and new phenomena. Students consider alternatives in constructing models based on analyses of the <i>different advantages and weakness</i> for explaining and predicting these alternative models possess.
2	<i>Students construct and use a model to illustrate and explain how a phenomenon occurs</i> , consistent with the evidence about the phenomenon. Students view models as a <i>means of communicating their understanding of a phenomenon</i> rather than a tool to support their own thinking.
1	<i>Students construct and use models that show literal illustrations of a single phenomenon.</i> Students do not view a model as tool to generate new knowledge, but do see models as <i>a means of showing others what the phenomenon looks like.</i>

Exemple 3:

Força i Moviment

(Alonzo y Steedle, 2009)

TABLE 2
Initial Version of the Force and Motion Learning Progression

Level	Description
5	Student understands that the net force applied to an object is proportional to its resulting <i>acceleration</i> (change in speed or direction), and that this force may not be in the direction of motion. Student understands forces as an interaction between two objects.
4	Student understands that an object is stationary either because there are no forces acting on it or because there is no <i>net</i> force acting on it. However, student may have misconceptions related to a belief that the applied force is proportional to an object's speed or motion (rather than its acceleration). Student can use phrases such as "equal and opposite reaction" to justify the existence of no net forces but may not understand this as an interaction. Common Errors: • Motion is proportional to the force acting. • A constant speed results from a constant force. • Confusion between speed/velocity and acceleration.
3	Student recognizes that forces are not <i>contained</i> within moving objects; however, student believes that motion implies a force in the direction of motion and that nonmotion implies no force. Common Errors: • Forces are associated only with movement. • Forces are viewed as causing things to move but not causing things to stop. • If there is motion, there is a force acting. • If there is no motion, then there is no force acting. • There cannot be a force without motion. • When an object is moving, there is a force in the direction of its motion.
2	Student recognizes that forces can be caused by nonliving things; however, student may believe that forces reside in within moving objects. Common Errors: • A moving object has a force within it that keeps it going. • A moving object stops when its force is used up.
1	Student understands forces as a push or pull, but believes that only living or supernatural things can cause forces. Common Errors: • Forces are caused by living things. • Forces are associated with physical activity or muscular strength. • Weight, motion, activity, and strength are important in determining an object's force.
0	No evidence or way off-track

Ex 4: Estructura de la matèria

(Talanquer 2009)

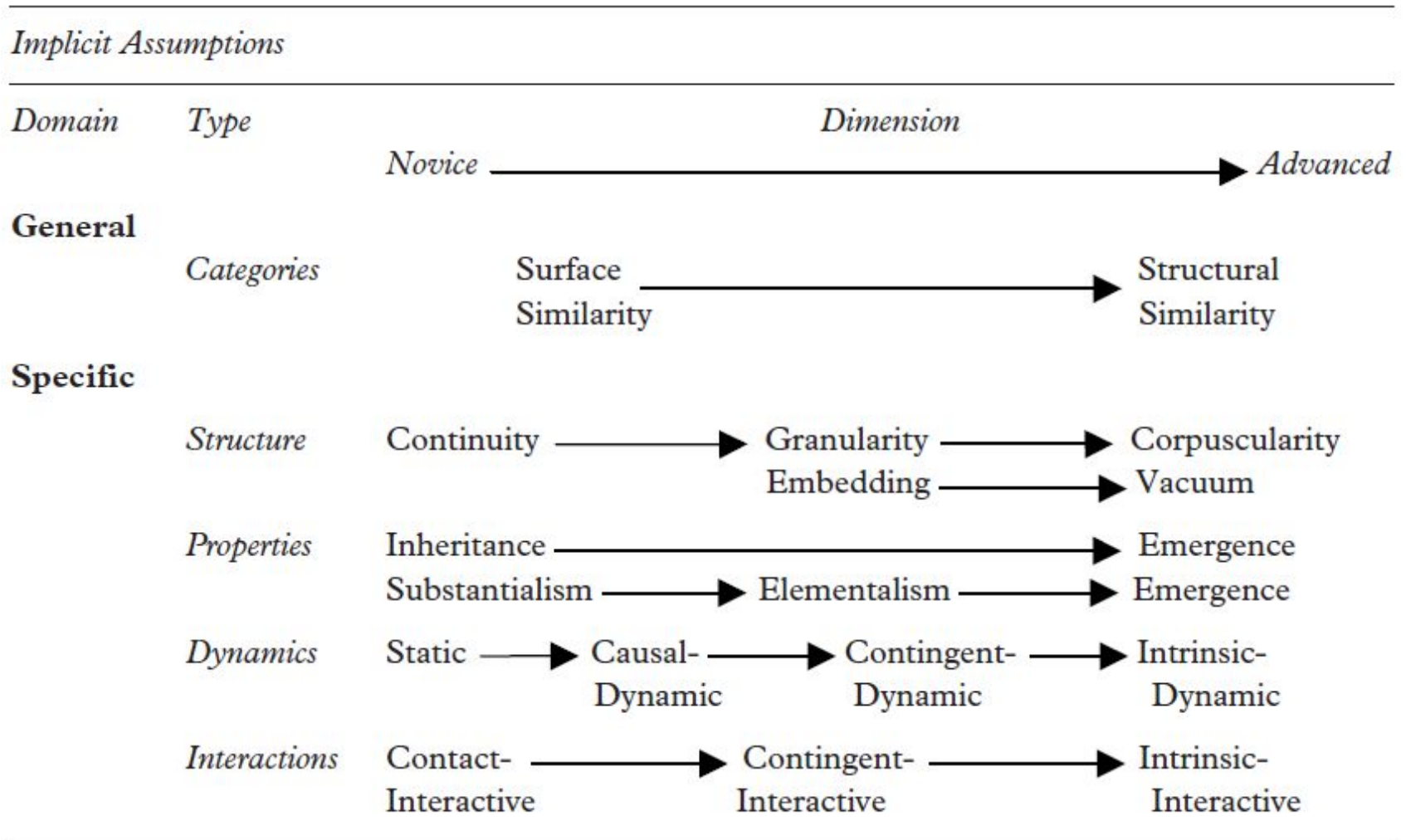


Figure 1. Implicit assumptions that constrain students' thinking about the structure of matter

Example 5: Energia

(Neumann et al 2013)

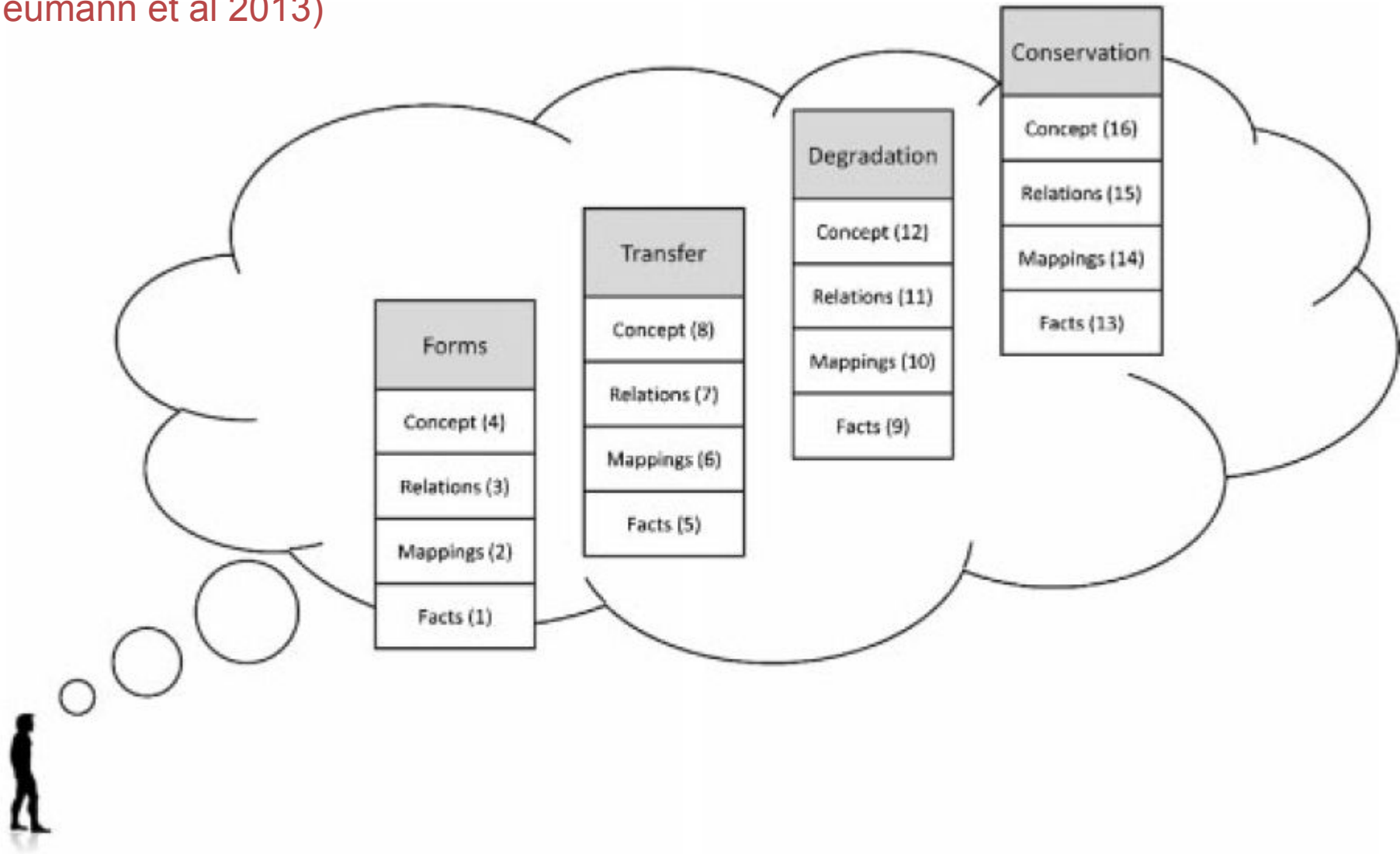


Figure 1. Initial learning progression of energy (visualization according to Wilson, 2009).

Ex 6: Evolució

(Lehrer und Shchauble,
2012)

Constructs	Levels			
	Initial	Developing	Stabilizing	Emerging
Learning performances Individual Change	Select & represent relevant attributes of organism	Represent change as difference	Represent change as rate/changing rate	Describe covariation among two or more rates of change
Mediated by	Physical models, drawings, Venn diagrams	Remnant models (e.g., pressed plants), paper strips, measure	Tables, Cartesian system, linear function	Bivariate display, function (exponential, logistic)
Variability	Observe/inscribe difference	Construct measures of individual difference	Represent individual differences as distribution	Develop process models and evaluate model fit
	Drawings (e.g., morphology of soldier fly vs. beetle larvae), remnants	Tables, lists	Frequency displays, statistics	Chance models of distribution
Population Change	Identify attribute of a population	Produce qualitative description of change in attribute over time	Describe change over time for a single variable	Characterize population growth
	Measure	Counts, difference, table	Piece-wise linear change, distribution	Function
Ecology	Consider place and organism using analogy to humans	Relate place and organism by considering needs of organisms	Partition place and consider effects on survival and reproduction of organisms	Model interactions among organisms and place to predict stability and transition
	Narrative	Drawings, field guides (constructed by teachers, students)	Maps, sampling schemes, models of chance variation	Electronic tools for model construction

Ex 7: Atomic Structure

(Park et al 2009)

Level	Description of mental model	Student model	Aspect of variation	Scientific model
1	Student understands atom as an indivisible particle.	P1a	Indivisible entity	Particle model (P)
	Student understands atom as an indivisible particle but also differentiates atoms and molecules.	P1b		
2	In addition to understanding an atom as a particle, the student understands that atoms are composed of protons, neutrons, and electrons.	N1	Subatomic components	Nuclear model (N)
3	In addition to level 2, the student understands the compositional relationship between protons and neutrons in a central nucleus, and electrons circling the nucleus)	N2	Circular orbit	
4	In addition to level 3, the student understands orbits with some reference to natural forces (gravity, strong, weak, and electromagnetic forces) holding them together.	N3	Circular orbit held by forces	
5	In addition to level 4, the student understands multiple different levels of orbitals as set paths of electrons in circular shape	B1a	Multiple orbits (paths)	Bohr's model (B)
	A student describes the Nuclear model (with ref to forces) and multiple different levels of orbitals as set paths of electrons in different shape.	B1b		
6	A student describes the Bohr's model and distinguishes orbital levels/shapes by energy quantization in quantum theory.	B2	Energy quantization	
7	A student arranges electrons not on a set path but within a certain area/space (or electrons within orbitals).	Q1	Orbital as space	Quantum model (Q)
8	A student describes electrons in terms of a space consistent with the meaning of probability based on the "uncertainty principle."	Q2	Probable space	
9	In addition to level 8, the student understands the different shapes of orbitals (but the different shapes are not representationally connected).	Q3a	Different space shapes	
	In addition to level 8, the student understands the different shapes of orbitals (and the different shapes are representationally integrated).	Q3b		
10	A student understands the basic idea of quantum theory (probability, wave function, energy quantization, etc.) and the modern picture of the atom and explains the atom in	Q4	Space shapes by mathematical wave function	

Ex 8: Argumentar (Berland y McNeill 2012)

Dimension	Simple  Complex		
Instructional context	Question is closely defined with two-three potential answers		Question is open with multiple potential answers
	Data set is small	Data set is large	Students define data set
	Data set is limited to appropriate data		Data set includes both appropriate and inappropriate data
	Detailed scaffolds	Moderate scaffolds	No scaffolds
Argumentative product	Claims are defended	Claims are defended with evidence	Claims are defended with evidence, and reasoning.
	Counterclaims are NOT rebutted		Counterclaims ARE rebutted
	Claim addresses question asked		Claim addresses question asked with a causal account
	Component (i.e. evidence, reasoning, rebuttal) is appropriate.		Component (i.e. evidence, reasoning, rebuttal) is appropriate and sufficient.
Argumentative process	Claims are articulated, defended, questioned OR evaluated	Claims are articulated, defended, questioned, AND evaluated	Claims are articulated, defended, questioned, evaluated, and revised
	Student participation in argumentative discourse is prompted by their teacher	Teacher and students share responsibility for prompting the argument	Students spontaneously engage in argumentative discourse.

Figure 1. Three dimensions of the learning progression. The more darkly the cells are shaded, the more complex the students' work on that characteristic.

Ex 9: Avaluar (Talanquer 2014)

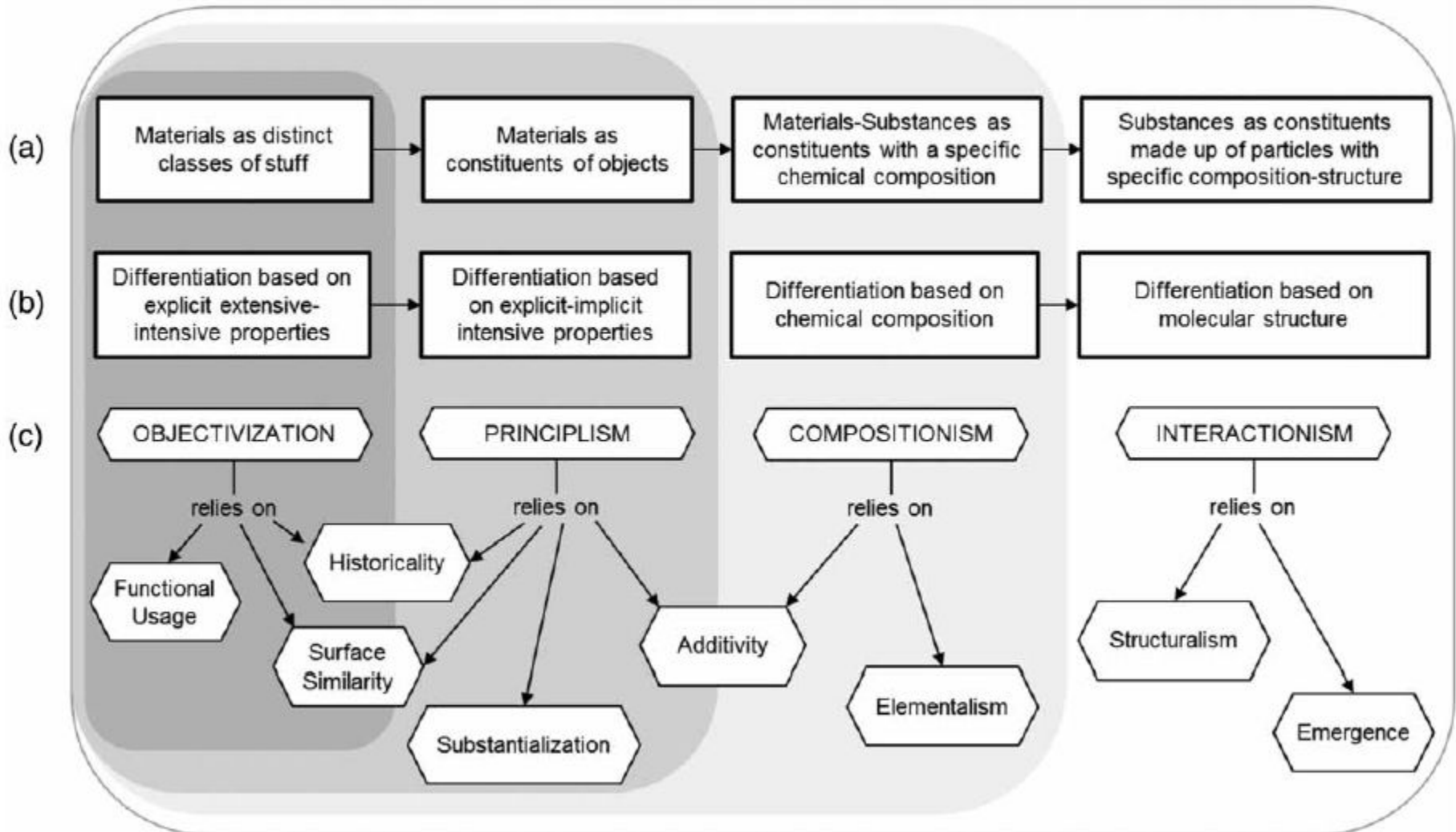
	Novato	Avanzado
¿Cómo se evalúa?	Uso limitado de estrategias de evaluación, en su mayor parte de carácter sumativo	Uso de preguntas informales con fines diagnósticos y formativos. Incremento en la diversidad de estrategias sumativas de evaluación
¿Para qué se evalúa?	Determinar si los estudiantes poseen el conocimiento académico correcto	Identificar lo que los estudiantes saben y no saben, para corregir problemas de aprendizaje
¿Cómo se interpretan los resultados?	Interpretaciones vagas y generales sobre el conocimiento y actitudes de los estudiantes	Interpretaciones específicas, pero no siempre productivas, sobre el conocimiento y formas de pensar de los estudiantes

Tabla 4. Progresiones comunes en modos de pensar sobre la evaluación de la disciplina

Ex 10: Identitat química (Ngai et al 2014)

NOVICE

ADVANCED



Ex 11 Rúbrica propietats acústiques materials

(Hernandez, Couso and Pinto 2014)

Table 7 Description of the stages of development of CM1 and students' answers

Stage of development of CM1	Description of students' models	Examples of students' answers
S1	Students explain sound attenuation in a material as the decrease in sound intensity level resulting from the distribution of sound in different components such as transmitted sound and reflected sound —LO1.1, LO1.2 & LO1.5	[When sound emitted by loudspeakers reaches the walls of a disco] it bounces back and some goes outside the disco through the wall
S2	Students explain sound attenuation in a material as the decrease in sound intensity level resulting from the distribution of sound in different components such as transmitted sound and absorbed sound —LO1.1, LO1.3, and LO1.5	[The part of sound that is not transmitted through the walls] has been absorbed by the walls

S3

Students explain sound attenuation in a material as the decrease in sound intensity level resulting from the distribution of sound in different components such as transmitted sound, **reflected sound, and absorbed sound**—LO1.1, LO1.4, and LO1.5

[The part of sound that is not transmitted through the walls] has been reflected or absorbed by the walls

S4

Students explain sound attenuation in a material as the decrease in sound intensity level resulting from the **distribution of energy of sound** in different components such as the energy of transmitted sound, the energy of reflected sound, and the absorbed energy—LO1.1, LO1.4, LO1.5, and LO1.6

To increase sound attenuation in the neighbors' house, the energy and the intensity of sound must decrease. To do so, the walls need to be made of good sound absorbers and insulators so that absorption and reflection of sound are higher

Ex 12. Lectura crítica

(Document avaluar per aprendre, Neus Sanmartí, web xtec)

Activitat escollida per avaluar resultats d'aprenentatge: Lectura crítica d'un article periodístic i elaboració d'un text argumentant el propi punt de vista tenint en compte els coneixements científics apresos *(per avaluar components de les competències comunicativa — lectura crítica i argumentació—, d'autonomia personal — pensament crític— i en el coneixement i interacció amb el món físic — coneixement concret que ha d'utilitzar per comprendre la lectura i fonamentar l'argumentació pròpia—)*.

Criteris d'avaluació de realització	Criteris d'avaluació de resultats (a concretar en indicadors en funció del contingut de la lectura i el curs)			
	Nivell 3 (expert)	Nivell 2 (avançat)	Nivell 1 (aprenent)	Nivell 0 (novell)
Identificar les idees clau de l'article.	Identifica totes les idees clau, les interrelaciona i les expressa amb paraules pròpies, d'una manera pertinent.	Identifica algunes de les idees clau i les expressa amb paraules pròpies, d'una manera pertinent.	Identifica algunes de les idees clau, a partir de reproduir literalment el que es diu al text.	Cita idees no pertinents.
Identificar el propòsit de l'autor, les suposicions que fa i el seu punt de vista.	Identifica, justificant-ho i de manera coherent, el punt de vista de l'autor, les suposicions que fa i que el seu propòsit és informar i crear polèmica.	Identifica, sense justificar-ho, el punt de vista de l'autor, algunes de les suposicions que fa i que el seu propòsit és informar i crear polèmica.	S'intueix que identifica el punt de vista de l'autor i les suposicions que fa reproduint frases literals del text. Creu que l'únic propòsit és informar.	Escriu idees irrelevantes o que no es poden inferir a partir del text.

A tenir en compte per a DEIA

Haureu d'incloure a les vostres SAs un apartat amb els següents punts sobre les idees clau de la unitat:

1. Què s'ha fet en **cursos anteriors** o etapes anteriors?
2. Quines **concepcions alternatives** s'han descrit a la recerca sobre el tema?

Feu servir cercadors d'internet amb la paraula clau “misconception”, “prior knowledge”, “ideas previas”, “concepciones alternativas” + el tema o concepte

3. Què s'haurà de fer en **cursos posteriors** a partir d'elles?

Exemple per a una idea clau sobre l'energia:

1. L'energia és allò que serveix perquè nosaltres i les màquines funcionem. (primària)
2. L'energia d'una pila desapareix quan aquesta es gasta. (Misconception)
3. L'energia es manifesta de diferents formes però no es crea ni es destrueix; es transforma i es transfereix i quan això passa sempre hi ha una part que es degrada, deixa de ser-nos útil. (2n - 3r ESO)
4. Càlculs d'energies fent servir el principi de conservació de l'energia mecànica. (4t ESO)

The END

1) Teniu el meu contacte:

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2) Dubtes sobre presentació i tasca al fòrum perquè tothom pugui llegir les respostes que us doni.

3) Gràcies per arribar fins aquí i us recordo el meu lema... ;)

“Per ensenyar ciències a la Naya,
has de saber molt de ciències,
molt de didàctica
i molt de la Naya i els seus companyxs”